Essential Eight Maturity Model - Summary

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Assessment Task 3 - Goal 2 Evidence

What is it?

The Essential Eight is a set of eight baseline mitigation strategies developed by the Australian Signals Directorate (ASD), drawn from its broader Strategies to Mitigate Cyber Security Incidents. The Essential Eight Maturity Model, published in June 2017 and updated regularly since, gives organisations a way to measure how well they have implemented on those strategies. It is based on ASD's experience in producing cyber threat intelligence, responding to incidents, conducting penetration testing and assisting organisations with implementation (ASD, n.d.-a).

The current version is the November 2023 update. That update focused on balancing patching timeframes, increasing adoption of phishing-resistant multi-factor authentication, managing cloud services, and incident detection and response for internet-facing infrastructure (ASD, 2023).

Strategy	What it does	Purpose
Application control	Allows only approved programs to run	Prevent malware delivery and execution
Patch applications	Applies security fixes to applications promptly, and removes unsupported software	Prevent malware delivery and execution
Configure Microsoft Office macro settings	Blocks macros from the internet and untrusted sources	Prevent malware delivery and execution
User application hardening	Disabled unneeded features in browsers and other applications	Prevent malware delivery and execution
Restrict administrative privileges	Limits who holds admin access and what they can do with it	Limit the extent of an incident
Patch operating systems	Applies operating system security fixes promptly and replaces unsupported versions	Limit the extent of an incident Limit the extent of an incident
Multi-factor authentication	Requires a second factor beyond a password	Limit the extent of an incident

Regular backups	Backs up important data, software and settings, and tests restoration	Recover data and maintain availability

In this case, the grouping is essential within the list. Four strategies prevent unwanted code from running, while three limit how far they can move if they do. At the same time, the last ensures data is saved, allowing the organisation to recover in catastrophic circumstances. No single strategy listed here is sufficient in isolation, proving how they are designed to be implemented together.

The maturity levels

ASD defines four maturity levels, from Maturity Level Zero to Maturity Level Three (ASD, n.d.-a).

Maturity Level	Meaning
Level Zero	Weaknesses exist in the organisations 's overall security posture
Level One	The organisation is protected against opportunistic adversaries using widely available tools and techniques
Level Two	The organisation protects against adversaries willing to invest more time and use more effective tools
Level Three	The organisation is protected against adaptive adversaries who are less reliant on public tools and will actively target weaknesses

Two rules govern how the levels are applied. First, organisations should identify a target maturity level suited to their own environment and threat profile. This is more efficient that assuming Level Three is always required (ASD, n.d.-b). Second, due to strategies nature to complement each other, organisations should reach the same maturity level across all eight before moving any of them higher (ASD, n.d.-a). In practice this means an organisation's maturity is set by its weakest strategy, not its strongest.

The model is also explicit about its limits: it is a minimum set of preventative measures, it will not mitigate every threat, and organisations may need additional controls depending on their environment (ASD, n.d.-a)

Why this matters for a project manager

The Essential Eight is traditionally discussed as a security team's concern, within an organisation that has adopted it. However, the Eight principles have controls arrive on every IT project as a requirement.

For example, a project that deploys new software must account for application control and patching from the beginning. A system with administrative accounts must fit the privilege restriction, meaning handling sensitive data needs a tested backup and restore

path before each go live. Each of these affect project scope, schedule and cost, where treating them as a late-stage sign-off items can cause these projects take longer than anticipated.

By understanding the model, project managers can surface those requirements during planning, when they are cheap and accounted for, rather than a security review, which can result in additional unexpected costs and time.

References

Australian Signals Directorate. (2023, November 27). *November 2023 Essential Eight Maturity Model update*. https://www.cyber.gov.au/about-us/view-all-content/news-and-media/november-2023-essential-eight-maturity-model-update

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